

$$\begin{aligned}
 1. \quad F &= CA \oplus B + AB(A + AB + AC + (AB' + A'C \odot B)) \\
 &= CA \oplus B + AB(A + AB + AC + (AB' + A'(B'C' + BC))) \\
 &= CA \oplus B + AB(A + AB + AC + AB' + A'B'C' + A'BC) \\
 &= CA \oplus B + AB(A(1 + B + C + B') + A'B'C' + A'BC) \\
 &= CA \oplus B + AB(A + A'B'C' + A'BC) \\
 &= CA \oplus B + ABA + ABA'B'C' + ABA'BC \\
 &= CA \oplus B + AB \\
 &= C(A'B + AB') + AB \\
 &= CA'B + CAB' + AB \\
 &= B(\underbrace{A'C}_{\overline{y-z}} + \underbrace{A}_{\overline{x}}) + CAB' \\
 &= B(\underbrace{A+A'}_1)(A+C) + CAB' \\
 &= BA + BC + CAB' \\
 &= BC + AB + CAB' \\
 &= BC + A(\underbrace{B + \frac{B'C}{\overline{y-z}}}_{\overline{x}}) \\
 &= BC + A(B+B')(B+C) \\
 &= BC + AB + AC
 \end{aligned}$$

$$\begin{aligned}
 2. \quad F(x, y, z) &= \Sigma(1, 3) \\
 &= x'y'z + x'yz
 \end{aligned}$$

